It all started with synthetic nucleic acids

Dr. Har Gobind Khorana was an Indian American biochemist and molecular biologist who shared the 1968 Nobel Prize in Physiology or Medicine with Marshall W. Nirenberg and Robert W. Holley for deciphering how nucleotides in nucleic acids control the synthesis of proteins. That discovery helped define one of the central principles of molecular biology: that information encoded in nucleic acid sequence could be translated into biological function.

For Khorana, the painstaking process of synthesizing short DNA and RNA sequences may have seemed slow, repetitive, and technically unforgiving. Yet that disciplined chemical work helped reveal something profound. Nucleic acids were not only the molecules of heredity; they could be constructed, modified, and used as experimental tools. In the mid-1950s, working in the laboratory of Sir Alexander Todd, Khorana pioneered the use of synthetic DNA and RNA to address fundamental biological questions. He showed that synthetic nucleic acids could serve as precise templates for DNA and RNA polymerases and helped establish an experimental framework that would later become central to synthetic biology, biotechnology, and therapeutic design.

RNA therapeutics begins here: once nucleic acids could be synthesized, they ceased to be mysteries of nature and became tools of design. They became molecules that could be designed, sequences defined, their chemistry altered, their stability improved, their interactions studied, and their biological effects directed toward specific genes, transcripts, proteins, cells, and tissues.

In the 1960s, Dr. Bob Letsinger, working with Dr. V. Mahadevan, developed one of the first practical methods for synthesizing DNA on an insoluble polymer: the phosphotriester method. Solid-phase synthesis transformed nucleic acid chemistry from a laborious solution-phase process into a more scalable and programmable technology. The efficiency of this approach helped enable the creation of the first bioengineered genes. Somatostatin, human insulin, and human growth hormone became early products of recombinant DNA technology and helped establish Genentech as one of the first biotechnology companies built on the idea that biological information could be converted into medicine.

Building on this foundation, Professor Marvin Caruthers advanced the field in the 1970s by developing nucleoside phosphoramidite chemistry for the solid-phase synthesis of DNA and RNA. This chemistry allowed oligonucleotides to be produced with precision, reproducibility, and scalability. It remains a foundational method in modern nucleic acid synthesis. At nearly the same time, the importance of oligonucleotide backbone chemistry was becoming clear. Eckstein and Gindl showed that replacing a non-bridging oxygen in the phosphate backbone with sulfur, generating phosphorothioate linkages, could preserve hybridization and polymerase compatibility while improving resistance to nuclease degradation. This observation established a principle that continues to shape the field: nucleic acids can be chemically modified to improve their pharmacological properties without necessarily eliminating their biological function.

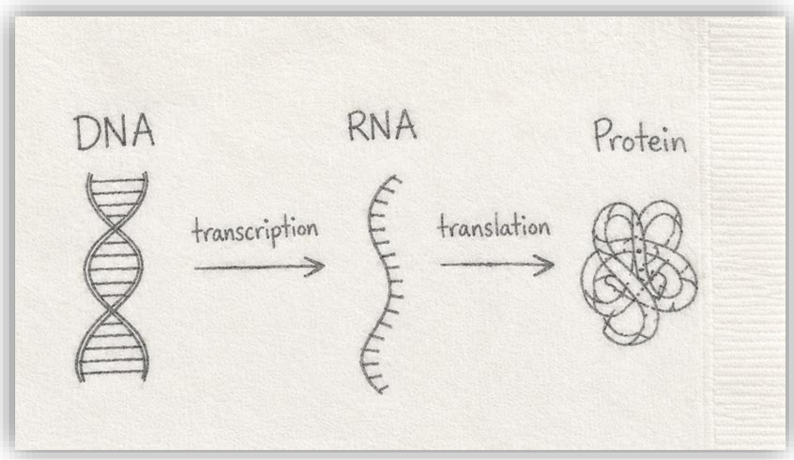
This principle is at the heart of RNA therapeutics. Native RNA is fragile, transient, highly charged, and rapidly degraded by nucleases. These properties make RNA a beautiful molecule for biology, but a difficult molecule for medicine. Therapeutic RNA technologies therefore depend on chemistry. Backbone modifications, sugar modifications, nucleobase substitutions, terminal conjugates, stereochemical control, lipid encapsulation, and ligand-directed delivery systems all exist to solve the same basic problem: how to preserve the informational power of nucleic acids while making them stable, deliverable, selective, and safe enough to function as drugs.

The first major therapeutic logic to emerge from synthetic oligonucleotide chemistry was antisense inhibition. Dr. Joseph Walder, founder of Integrated DNA Technologies, demonstrated that antisense oligonucleotides could inhibit gene expression either by blocking translation or by recruiting RNase H enzymes to cleave target RNA. This linked sequence-specific hybridization to enzymatic RNA degradation and helped define what is now known as the gapmer antisense oligonucleotide. In a gapmer, chemically modified nucleotides at the ends of the molecule improve stability and binding affinity, while a central DNA-like region supports RNase H recruitment. The result is a programmable drug that can bind a selected RNA transcript and direct its destruction.

A second major therapeutic logic emerged with the discovery of RNA interference. In the 1990s, Dr. Andrew Fire and Dr. Craig Mello showed that double-stranded RNA could induce potent and sequence-specific gene silencing. This discovery revealed an endogenous cellular pathway in which double-stranded RNA is processed into small interfering RNAs, which then guide the RNA-induced silencing complex to complementary messenger RNAs. The guide strand provides specificity, while the cellular machinery provides catalytic silencing. RNA interference transformed the idea of gene knockdown from a synthetic antisense strategy into a programmable use of natural cellular regulation.

These two mechanisms, RNase H-mediated degradation and RISC-mediated degradation, became pillars of RNA-targeting therapeutics. Both use base pairing to identify a target transcript. Both can be designed from genetic sequence. Both convert molecular recognition into reduction of gene expression. Yet they differ fundamentally in chemistry, structure, intracellular compartment, protein machinery, delivery requirements, duration of action, and pharmacological behavior. Understanding these differences is essential because “RNA therapeutic” is not a single mechanism. It is a family of molecular strategies that share the use of nucleic acid information while relying on distinct biological pathways.

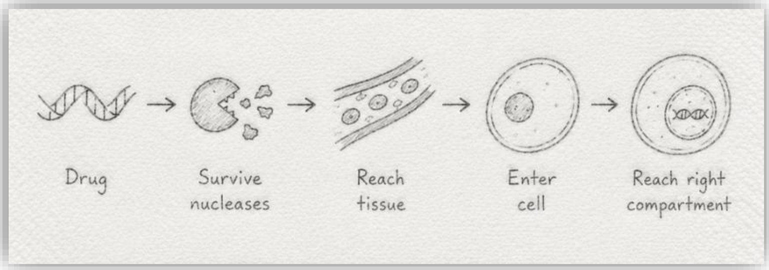
Alongside these degradation-based approaches, steric-blocking oligonucleotides opened a different therapeutic possibility. Rather than recruiting an enzyme to destroy RNA, these molecules bind RNA and physically block access by proteins, ribosomes, spliceosomal components, microRNAs, or other regulatory factors. Phosphorodiamidate morpholino oligomers, developed by James Summerton and colleagues, were an important advance in this area. In morpholinos, the ribose sugar is replaced by a morpholine ring and the charged phosphodiester linkage is replaced by a neutral phosphorodiamidate backbone. This architecture confers strong nuclease resistance and allows the molecule to bind RNA without activating RNase H.



Steric-blocking chemistry made it possible to think of RNA not only as a transcript to be inhibited, but as a substrate to be remodeled. By blocking splice sites, splice enhancers, splice silencers, polyadenylation signals, upstream open reading frames, microRNA binding sites, or translational regulatory elements, oligonucleotides can alter how an RNA is processed and interpreted. This can increase productive protein output, suppress toxic isoforms, induce exon skipping, restore a reading frame, promote nonsense-mediated decay, or shift the balance between alternative transcript isoforms. In this way, RNA therapeutics move beyond simple gene silencing and enter the domain of transcript engineering.

The central dogma describes the flow of biological information from DNA to RNA to protein. RNA therapeutics intervene in the middle of that flow. They do not cause changes to DNA. Instead, they exploit the regulatory space between genome and proteome. This gives RNA therapeutics unusual flexibility. A drug can silence a toxic gain-of-function transcript, reduce a disease-associated protein, restore splicing, increase expression from a healthy allele, modulate a pathway by shifting isoform balance, or introduce a transient protein-coding message into a cell.

The clinical landscape now reflects this breadth. Antisense oligonucleotides have been approved for neurological, metabolic, and rare genetic diseases. siRNA therapeutics have validated durable gene silencing, especially in the liver. Splice-switching oligonucleotides have changed the treatment of diseases such as spinal muscular atrophy and Duchenne muscular dystrophy; in SMA, this shift has been profound. Spinal muscular atrophy is no longer a leading cause of inherited infant death in the United States, following the approval of three genetic therapies since 2016. Aptamers have shown that structured nucleic acids can bind protein targets with antibody-like specificity. mRNA vaccines can direct transient protein expression at a global scale. Short activating RNAs, circular RNAs, self-amplifying RNAs, and microRNA-directed approaches further expand the field from gene silencing toward gene activation, protein expression, immune modulation, and programmable regulation.



mRNA therapeutics represent anotherconceptual approach where instead of targeting an endogenous RNA transcript, synthetic RNA introduces genetic instructions into cells. Once delivered to the cytoplasm, this RNA is translated into protein. This makes mRNA useful for vaccination, protein replacement, immunotherapy, genome editing, and cellular reprogramming. The success of COVID-19 mRNA vaccines brought this modality into public view, but the underlying technology rests on decades of work in cap biology, and sequence design, codon optimization, nucleoside modification, innate immune recognition, lipid nanoparticle delivery, and manufacturing science. Modern mRNA therapy is not simply copied from a gene; it is designed from end to end.

Across all of these modalities, delivery remains one of the defining challenges. A nucleic acid drug must survive extracellular nucleases, avoid rapid clearance, distribute to the relevant tissue, enter the appropriate cell type, escape or productively traffic through endosomal compartments, and reach the intracellular location where its mechanism operates. For siRNAs, this often means cytosolic access and RISC loading. For splice-switching ASOs, it often means nuclear delivery. For mRNA, it means cytosolic translation. For RNA editing, it may mean coordinated localization of the editing oligonucleotide, target RNA, and endogenous editing enzyme. The same sequence can be potent in a dish and ineffective in vivo if it cannot reach the right compartment.

The liver has become the most tractable organ for systemic RNA therapeutics because of efficient hepatocyte uptake mechanisms, particularly GalNAc conjugation for siRNAs and antisense oligonucleotides, as well as lipid nanoparticle delivery. Other tissues remain more difficult. The central nervous system often requires intrathecal administration. The eye can be accessed by intravitreal injection. Muscle, lung, immune cells, and solid tumors each present different barriers of anatomy, circulation, uptake, and intracellular trafficking. For this reason, delivery is not a secondary formulation problem. It is part of the drug's mechanism.

Pharmacology is equally distinct. RNA therapeutics do not behave like small molecules or antibodies. Plasma concentration may fall rapidly while tissue activity persists for weeks or months. Total tissue concentration may not predict pharmacodynamic effect if most of the drug is trapped in nonproductive compartments. Chemical modifications may increase stability but alter protein binding, biodistribution, potency, toxicity, or clearance. A gapmer ASO, a fully modified splice-switching ASO, a GalNAc-siRNA, a lipid nanoparticle-formulated mRNA, and an aptamer each has a different relationship between dose, exposure, tissue distribution, intracellular availability, target engagement, and biological response.

Manufacturing also shapes the field. Nucleic acid therapeutics require careful control of sequence identity, purity, protecting-group removal, chain length, stereochemistry, conjugation, encapsulation, residual solvents, truncated products, double-stranded impurities, endotoxin, innate immune contaminants, and analytical characterization. For mRNA, additional considerations include capping efficiency, poly(A) tail length, nucleoside incorporation, double-stranded RNA impurities, lipid nanoparticle composition, particle size, encapsulation efficiency, and stability. These are not merely technical details. They determine whether a molecule can become a reproducible, regulatable, and scalable medicine.

RNA therapeutics sit at the intersection of chemistry, biology, medicine, and engineering. Their power comes from programmability, but programmability alone is not enough. A sequence must become a molecule. A molecule must become a drug. A drug must reach a tissue. Tissue exposure must become productive intracellular activity. Each step requires design.

The future of RNA medicine will likely be defined by increasing precision over each of these steps. Improved chemical modifications will extend stability and tune protein interactions. Better conjugates and delivery vehicles will expand tissue reach beyond the liver. Computational tools will improve target selection, structure prediction, off-target analysis, and sequence optimization. RNA editing and activation may broaden the field from suppression of disease genes to restoration, correction, and controlled enhancement of gene function. Personalized and individualized RNA medicines may make it possible to design therapies for mutations, splice defects, or rare diseases that were previously unreachable by conventional drug discovery.